

Enterprise Digital Infrastructure

Monitoring



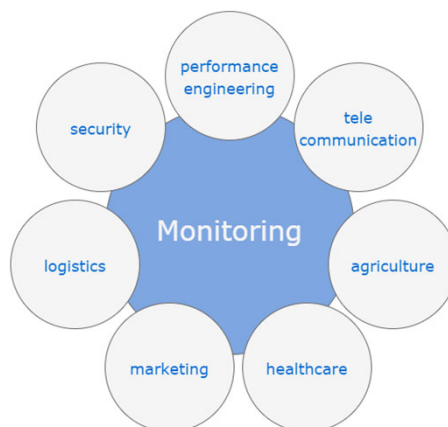
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Monitoring

Monitoring, i.e., the process of collecting measurements, is a key activity in many application domains because it captures the actual behavior of the phenomenon under investigation and provides its quantitative and qualitative descriptions



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Performance monitoring

Performance monitoring is a fundamental component to properly manage infrastructures and services

Main purposes of monitoring activities:

- Understand the behavior and performance of services and infrastructures
- Assess their health
- Anticipate potential outages and SLA violations
- Diagnose performance issues when something goes wrong
- Proactively address and mitigate performance issues
- Investigate user behaviors

Data being monitored is of interest of many diverse stakeholders, e.g., end users, software developers, webmasters, service providers, network operators, cloud operators

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Basic concepts

Monitoring denotes the *process* of measuring “something” by means of a *device* or a measurement instrument/tool in a standard *measurement unit*

The result of the process is a number or a set of numbers, i.e., *measurements*, that provide a *quantitative* assessment of the measured object

Measurements must be specific of the components under investigation

The *accuracy* of the process and the *precision* of the measurement tools are *necessary* ingredients for the validity of the measurements, although not *sufficient*

A correct analysis and interpretation of the measurements are required before drawing any meaningful conclusion

Solid domain knowledge is required and a multi-disciplinary approach

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Monitoring project workflow

The plan, design and setup of a monitoring project are formulated as a workflow consisting of multiple *interdependent* steps referring to the diverse aspects to be considered

The following questions need to be answered:

- WHY: definition of the *objectives* of the project
- WHICH: definition of the *target(s)* of the project
- WHAT: definition of the *attributes* to be measured
- WHERE: definition of the *vantage point(s)*
- HOW: definition of the *measurement process*
- WHEN: definition of the *timing*

Once measurements have been collected, they need to be *summarized* to build *models* representative of the properties and behavior of the target

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WHY

This step refers to the definition of (clear) objective(s) of the monitoring project

Possible objectives:

- to understand how a technological infrastructure or a service actually work
- to discover bugs (*debugging*)
- to detect and diagnose inefficiencies (*performance debugging*)
- to assess the status of the technological components (*predict failures*)
- to measure the usage of infrastructure & services (*accounting & billing*)
- to detect anomalies (*security*)
- to understand the behavior and activities of the users (*profiling*)

Data is the gold of the digital era

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WHICH

This step refers to the definition of the *target*, i.e., System Under Test (SUT), of the monitoring project

The choice depends on the objectives of the project (i.e., why)

Possible targets:

- *Technological infrastructure* at different levels (not to be considered in isolation)
 - computing infrastructure
 - storage infrastructure
 - communication infrastructure
 - software components
- *Workload*
- *Network traffic*
- *Users* (e.g., GDPR compliance)

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WHAT

This step refers to the definition of the *attributes*, i.e., properties of the target

These choices also depend on the usage of the measurements

- *Offline*
- *Online/on the fly*

Possible attributes:

- number of processes, RAM usage, CPU usage, number/types of instructions, number of cache misses, number of I/O operations
- application processing time, number of DNS queries, number/type of DB transactions, number/types of HTTP transactions, user navigation profile
- throughput, latency, jitter, error rate, number of dropped packets, protocol types
- GPS coordinates, energy, battery usage, temperature

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WHERE

This step refers to the definition of the *vantage point(s)*, i.e., observation point(s), where measurements about the target are collected

Any component of an infrastructure could be chosen as a vantage point

The choice has to take into account two important aspects: *feasibility* AND *impact*

Feasibility refers to the *permissions/authorizations* required to monitor

Impact refers to *side effects* such as *perturbations*, *intrusiveness* and *overhead* introduced by monitoring activities (e.g., monitoring frequency, length of monitoring interval, amount of collected data)

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HOW

This step refers to the definition of the *measurement process* that includes the choice of the *techniques* and *tools* to be used to collect measurements

The choice is between two categories:

- *Active* techniques aimed at collecting measurements on the target component under a *controlled* workload/traffic

artificial workload/traffic & real infrastructure

- *Passive* techniques aimed at collecting measurements on the target component while operating under its *real* workload/traffic

real workload/traffic & real infrastructure

The measurement process usually leverages (properly customized) open source or proprietary *tools*

These tools can be used in isolation or in combination

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WHEN

This step refers to the definition of the timing of the monitoring project, i.e., when measurements are to be collected

Possible choices:

- Continuously
- Periodically
- Occasionally

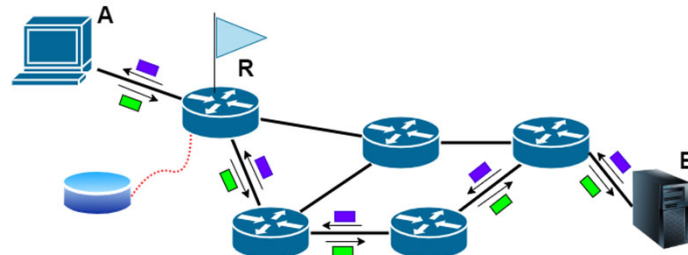
In general, it is recommended to consider monitoring as a *systematic* activity rather than a sporadic activity

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Example of monitoring scenarios



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Active monitoring

Active monitoring belongs to the broad category of *load* and *traffic generators*

It is aimed at analyzing the properties and assessing the efficiency of a real infrastructure under a “controlled” (artificial) workload/traffic

Main phases:

- definition of the characteristics of the artificial load/traffic (i.e., *benchmark*)
- generation of artificial load or traffic on the real infrastructure chosen as target
- monitoring the performance and behavior of the target

The complexity of the generators varies from simple commands/utilities that are part of the operating systems, to a large variety of speed testing tools and complex benchmarking frameworks

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Active monitoring (cont'd)

The target can be *shared* or *dedicated*

The artificial load is generated according to some predefined characteristics and behaviors

To increase accuracy of the experiments and ensure measurement quality, it is necessary to

1. Generate a *realistic/representative load* (consisting of multiple requests/packets)
2. Repeat the experiments *multiple times* and discard inaccurate samples

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Monitoring project: examples

Setup a monitoring project aimed at evaluating the health of a communication infrastructure

Simple commands for active monitoring:

- **ping**
- **tracert**
- **mtr**
- **hping3**
- **nmap**
- ...

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Active monitoring: **ping**

ping is a simple active monitoring tool part of the operating systems aimed at managing, measuring, testing and debugging network configurations

The tool relies on the Internet Control Message Protocol (ICMP), a network layer protocol aimed at error reporting and diagnosis [RFC 792 released in 1981]

ICMP supports 11 types of messages sent as IP datagrams

Ping generates packets from a sender (i.e., *vantage point*) towards a *target*

Each packet consists of ICMP and IP headers (8+20 bytes) and variable payloads (pad bytes to fill the packet)

Ping is based on two ICMP messages: ECHO msg (type 8) that elicits an ECHO REPLY message (type 0)

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Active monitoring: **ping** scenario

Why:

- Debugging network configuration

Which:

- Server B

What:

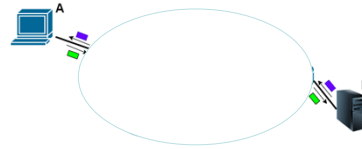
- Reachability of the target
- Instantaneous RTT between vantage point and target
- Packet loss rate

Where:

- Computer A

How:

- **ping**



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